Class 1, Week 8: Your first model: baseline model, linear regression and prediction error

Lecturer guide for CML3112, CML3113 and CML3111. Keep it private.

The class in one paragraph

How can a computer predict cooling load from load and temperature, and how do we know it is any good?

Welcome to the second half of the semester. Today is your first full machine learning model. We use the building data and we predict cooling load. You have spent seven weeks learning to read, clean and chart data. Now that work pays off, because a model is only as good as the rows we give it. The ideas come first, then the lab.

When each cohort meets

Cohort	Programmes	Date	Time and room
CML3112	Civil Engineering and Architecture	Tuesday 3 November 2026	11:00 to 14:00, Room 106
CML3113	Mining and Petroleum Engineering	Friday 6 November 2026	08:00 to 11:00, Room 105
CML3111	Electrical, Mechatronics, Robotics and Communications	Friday 6 November 2026	11:00 to 14:00, Room 115

Before the class

- Open the deck in Slide Show mode and the lab notebook in Colab before anyone arrives. Run the notebook from the top once so you know it works on your account.
- Put the lab notebook and engineering_data.csv in the shared class folder and post the link. Every learner needs both files in Colab, in the Files panel.
- Write the "check your numbers" table on the board before the lab (numbers for your cohort are in the table below).
- The graded assignment notebook (W08_student.ipynb) is not used in class. Do not open it.

The clock

Tuesday cohort (CML3112): the first full teaching of this class. Friday cohorts (CML3113, CML3111): the concepts were taught last Friday, so the deck is used as a recap.

CML3112, Tuesday

Starts	Minutes	What happens
0:00	10	Recall from Weeks 1 to 7
0:10	35	Concepts, part 1: data, baseline model and prediction error
0:45	30	Concepts, part 2: learning, held-out data and models
1:15	15	Break
1:30	80	Lab: your first model, in the notebook
2:50	10	Close and what is assessed

CML3113 and CML3111, Friday

Starts	Minutes	What happens
0:00	10	Recall from Weeks 1 to 7

Starts	Minutes	What happens
0:10	35	Concepts: your first model, with pictures
0:45	55	Lab, part 1: prepare the rows
1:40	15	Break
1:55	55	Lab, part 2: three models and the model comparison table
2:50	10	Close and what is assessed

Slide map

The deck has two halves. Slides up to the divider are concepts: no laptops. After the divider come the lab slides and the notebook. You switch once.

Slide	Title	Part	Friday recap
1	Your first model: baseline model, linear regression and prediction error	Concepts	
2	Today in one view: what you will leave with	Concepts	
3	Your three hours: ideas first, then the lab	Concepts	
4	Three questions from last time	Concepts	
5	Answers to the three questions at the start of the class	Concepts	recap
6	Words we use today	Concepts	
7	A prediction turns past readings into a number you can act on	Concepts	
8	A glossary you will meet again in textbooks and documentation	Concepts	recap
9	Machine learning finds the rule from examples, so nobody writes it by hand	Concepts	recap
10	Every row is one reading: the inputs are features and the answer is the target variable	Concepts	recap
11	Before we model, we remove the rows we cannot trust	Concepts	recap
12	The baseline model comes first: predict the mean of the training targets every time	Concepts	recap
13	The baseline model as a formula: the mean of the training targets	Concepts	
14	Simple linear regression fits a line through the dots and beats the baseline model	Concepts	recap
15	Simple linear regression is a formula with two learned numbers	Concepts	
16	The prediction error (residual) is the actual value minus the predicted value	Concepts	recap
17	The prediction error of one row is the actual value minus the predicted value	Concepts	
18	Worked example with hypothetical numbers: four prediction errors give MAE 2.75 and RMSE 3.5	Concepts	recap
19	Three ways to turn many errors into one number: MAE, MSE and RMSE	Concepts	
20	Our own validation rows: the baseline model and simple linear regression	Concepts	recap
21	MAE and RMSE both measure prediction error in the target's own unit	Concepts	
22	Gradient descent trains a model by walking downhill on the loss, one small step at a time	Concepts	
23	Gradient descent, part 1: how steeply the loss slopes	Concepts	
24	Gradient descent, part 2: change w a little, downhill	Concepts	

Slide	Title	Part	Friday recap
25	On a tiny illustrative example, five steps almost reach the best slope	Concepts	
26	Multiple linear regression has three learned parameters: b0, b1 and b2	Concepts	
27	What one more unit of a feature does to the prediction	Concepts	
28	Hold out rows while the model trains, so the evaluation is fair	Concepts	
29	Imputation fills a missing value with the training median of its column	Concepts	recap
30	A decision tree learns a short list of yes or no questions about the features	Concepts	recap
31	max_depth is the largest number of questions a row can be asked	Concepts	
32	Model comparison: three models evaluated on the same validation set	Concepts	recap
33	A pipeline chains preprocessing and a model, and runs its steps in order	Concepts	
34	Read the workflow before you write it: create, fit, predict, score	Concepts	recap
35	Part 2: the lab	Lab	
36	The lab has five stages, and the notebook numbers its steps 0 to 8	Lab	
37	Check your numbers against these before you move on	Lab	
38	Four errors you will meet, what they mean and what to do	Lab	
39	How this class counts towards your coursework	Lab	
40	Save the notebook and keep your three MAE scores	Lab	

Friday recap: show the slides marked "recap" and pass quickly over the others in one sentence each. The learners met the ideas last Friday; the aim is to refresh them with the pictures before the lab.

Questions to ask, and a good answer

Ask	A good answer
Why is the baseline model the first model we build?	It sets the score to beat. A model that cannot beat the mean of the training targets has learned nothing useful.
Which is bigger, the MAE or the RMSE?	The RMSE is always at least as big, because squaring makes large errors count for more.
Why do we split by equipment group and not at random?	So the model is judged on equipment it has never met, which is the situation we face in real use.
Why must the imputer learn its median from the training set only?	If the validation set decided the fill, information from held-out data would reach the model (data leakage) and the score would be too kind.
Is the tree better because it is more complicated?	No. On these data multiple linear regression scores better on the validation set. The model comparison table decides, and when scores are close we prefer the simpler model (parsimony).
What does a positive error mean?	The actual value was higher than the prediction, so the model predicted too low.
What is the difference between a parameter and a hyperparameter?	A parameter is learned from data, such as the slope. A hyperparameter is a setting we choose before training, such as max_depth.
Is gradient descent what LinearRegression does inside?	My understanding is that LinearRegression uses a direct formula. Gradient descent is the same idea of walking downhill on the loss, used where no direct formula exists, such as in neural networks.

Ideas learners often get wrong

What a learner may say	What to say back
A model that is perfect on its training set is excellent	It may be overfitting: fitting noise in the training set. We always evaluate a model on held-out data.
Machine learning means the computer understands the problem	It finds a numerical rule from examples. It has no understanding, which is why we check its work.
More input columns always help	They can add noise. The lab step "only load_pct" tests this.
The test set is for choosing the best model	The validation set chooses. The test set stays held out until the end.

Walking the lab

Step	What to look for
Step 0 and 1: open and look	Shape (91, 5). Missing values: one in target and one in ambient_c. Anyone who sees an error here has the CSV in the wrong place.
Step 2: clean a copy	After removing repeats: 90. Rows left to model with: 88. If someone has 89 or 87 they ran a cell twice or changed the 100 and did not change it back.
Step 3: split	Rows 58, 15, 15. One missing value remains in the training features (ambient_c).
Step 4: baseline model	The same number for every row. The five-row table shows the arithmetic of error. Ask one learner to read a row aloud.
Step 5: linear regression	Intercept and two coefficients (b0, b1, b2 on the slides), both coefficients positive. The CHANGE task (simple linear regression, one feature) gives a larger MAE: the second feature helps.
Step 6: tree	Depth 4 gives the number on the board. The depth experiment shows the tree changing: park the explanation for Class 3.
Step 7 and 8: model comparison table and save	Three rows, multiple linear regression best. Insist on restart and run all before saving.

The numbers each cohort should see

Check	CML3112 (kW)	CML3113 (kWh/t)	CML3111 (deg C)
Rows in the file	91	91	91
After the three bad rows are removed	88	88	88
Training, validation, test rows	58, 15, 15	58, 15, 15	58, 15, 15
Baseline model, MAE	9.69 kW	2.71 kWh/t	14.85 deg C
Multiple linear regression, MAE	0.74 kW	0.67 kWh/t	0.78 deg C
Decision tree max_depth 4, MAE	3.39 kW	1.20 kWh/t	4.79 deg C
Multiple linear regression, intercept, coefficient of load_pct, coefficient of ambient_c	1.493, 0.447, 0.756	1.157, 0.120, 0.159	1.662, 0.695, 1.054
Multiple linear regression, RMSE	0.88 kW	0.81 kWh/t	0.92 deg C

If things go wrong

- If the concepts run over, skip gradient descent and the tiny example. Say: the computer improves the parameters of a model by small steps, and you will see it in a later class.
- If only thirty minutes of lab remain, do steps 0 to 5 and 7. Set step 6 (the tree) as practice for next time.

- If wifi fails in the room, use the offline set-up from Week 2: the notebook runs unchanged in VS Code.
- If several learners are stuck at the same cell, stop the room, run that cell on the projector and let them copy the fix.

After the class

- Share the lab notebook and engineering_data.csv again so absent learners can follow.
- Tell the class that nothing is submitted today. The graded part A follows the same steps with the assignment notebook.
- Note any question you could not answer and start the next class with it.

Speaker script

Every slide carries its spoken script in the PowerPoint notes (open Presenter View). The scripts are first person and are meant to be read or paraphrased. Pauses are marked by questions.